INCH-POUND
A-A-59941
12 December 2013

COMMERCIAL ITEM DESCRIPTION

CAPACITANCE STANDARD, FIXED

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE**.** This Commercial Item Description (CID) provides accurate reference values for the calibration of test measurement and diagnostic equipment for the development, repair and maintenance of navigational aids, radio frequency communications equipment and other measurement equipment. This CID lists the minimum performance, design, functional, environmental, reliability requirements and quality assurance provisions for the capacitance standard.

2. CLASSIFICATION. The capacitance standard covered by this CID is commercially available equipment that has factory installed options and/or may be modified to the extent necessary to meet the following description. The test set shall be Class 3, in accordance with MIL-PRF-28800.

3. SALIENT CHARACTERISTICS.

3.1 Item Description. This capacitance standard shall provide accurate reference values for the calibration of test measurement and diagnostic equipment for the development, repair and maintenance of navigational aids, satellite and radio frequency communications equipment and other measurement equipment. This standard shall be contained in an enclosure to maintain a stable environment and avoid RFI/EMI interference.

3.2 Capacitance Values. The capacitance standard shall have minimum and maximum values with tolerance of ±2.0% levels as show on table 1 Capacitance Value and Variability.

Comments, suggestions, or questions on this document should be addressed to: WR-ALC/404 SCMS/GUEEA, Robins AFB GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.,mil.

AMSC N/A FSC 6625

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Capacitor Values (Tolerance)	
10 pF	(+/-0.2 pF)
500 pF	(+/-10 pF)
900 pF	(+/-18 pF)
5000 pF	(+/-100 pF)
9000 pF	(+/-180 pF)

Table 1 Capacitor Values and worst case tolerance variation

3.3 Capacitor Types. The capacitance standard shall have non-polarized class 1 capacitors with dielectric material constant value between 15 and 100 with a delta percentage change of less than 0.4% from -55°C to 125°C (-67°F to 257°F).

3.4 Effective Series Resistance (ESR). The capacitance shall exhibit a maximum ESR equal or less than the absolute value of 2 ohms in alternate current measurement mode from wiring harness connector to connector, (see figure 1).

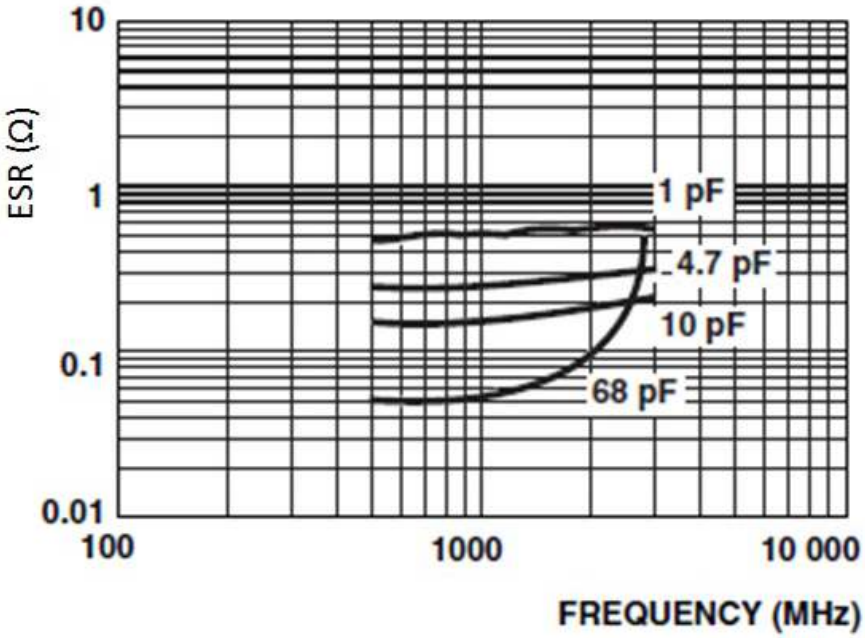


Figure 1 Effective Serial Resistance vs. Frequency

3.5 Direct Current Voltage Operation. The capacitance standard shall have low drop out voltage small signal fast recovery diodes connected in series with the capacitors. A terminal connector tapped, allowing for connectivity with or without the diode(s) in series as to allow for direct current and alternating current applications, shown in figure 2.

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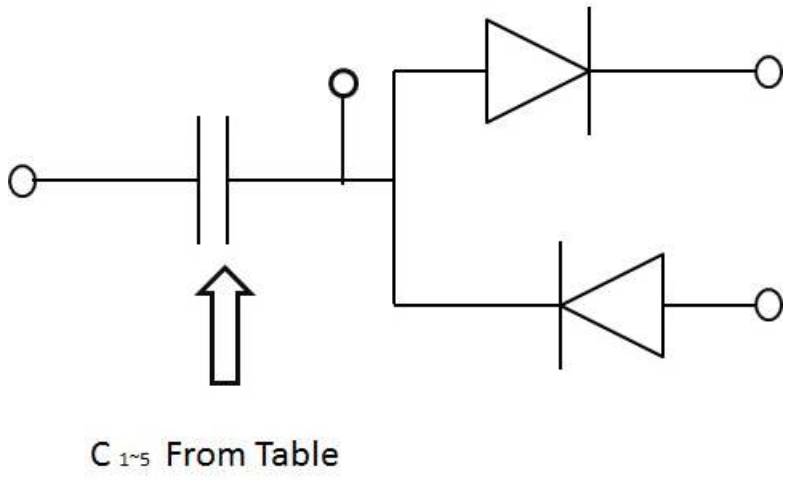


Figure 2 Equivalent Schematic Diagram

3.6 Working Voltage. The capacitors in the capacitance standard unit shall be devices rated at a voltage equal or greater than 75 volts in working direct current voltage, {$WV_{DC} \geq 75V$}.

3.7 Insulation Resistance (IR). The capacitance standard discrete devices shall exhibit a typical characteristic resistance (R_{CH}) ≥ to 6.3 Giga ohms or R x C ≥ 500 Ohms times, capacity in Farads after 120 seconds at working voltage (WV_{DC}).

3.8 Temperature Coefficient of Capacitance (TCC). The capacitance standard independent discrete devices shall have a typical TCC variability of ±60 ppm/°C from 55 °C to + 125 °C (131°F to 257°F).

3.9 Thermal Stability. The capacitance standard accuracy shall vary no more than ±2.0% of its value within the temperature range of 20°C to 30°C (68°F to 86°F) after tests in MIL-PRF-28800, section 4.5.5.1.1 has been performed for class 3 equipment classification (Per table 11 on page 41). The capacitance standard stability shall be within the values shown in table 2 at rated temperatures.

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Class	Temperature limits	
	Nonoperating equipment (°C)	Operating equipment (°C at percent RH)
1 [1]	-51 to 71	-40 to 71 at 5 to 60 -40 to 40 at 5 to 85 -40 to 30 at 5 to 95
2	-51 to 71	-10 to 55 at 5 to 45 -10 to 40 at 5 to 75 -10 to 30 at 5 to 95
3	-40 to 71	0 to 50 at 5 to 45 0 to 40 at 5 to 75 0 to 30 at 5 to 95
4	-40 to 71	10 to 40 at 5 to 75 10 to 30 at 5 to 95

Table 2 Operating Temperatures Values.

3.10 Design and Construction. The capacitance standard shall be used in both bench top, flight line and rack mounted environments. The capacitance standard is a commercial off the shelf item subject to commercial products regulatory standards. The capacitance standard shall comply with all applicable MIL-PRF-28800F sections. The capacitance standard shall be designed and constructed in accordance with all applicable European Union (EU) requirements in order to have the "CE" marking affixed.

3.11 System Safety. The system shall comply with all sections of the Underwriters Laboratories (UL), IEC/ISO standard 61010A and 61010B for electrical safety.

3.12 Sound Levels. The capacitance standard acoustic noise shall conform to sound level requirements on section 1 of ISO 7779 and sections 1 & 2 of ISO 8253.

3.13 Electrostatic Discharge (ESD). The design of the capacitance standard shall preclude equipment damage due to ESD (internal/external electrical arcing), protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking in accordance with MIL-STD-1686 sections 5.2 and 5.7.2 and MIL-HDBK-263 for guidance.

3.14 Design and Construction. The Frequency Domain Reflectometer shall be design and constructed for commercial use and shall meet the definition of Commercial Off-The-Shelf (COTS) items in FAR 2.101 The Frequency Domain Reflectometer shall be designed and constructed in accordance with all applicable European Union (EU) requirements in order to have the "CE" marking affixed, International Organization for Standardization (ISO), the International Electro-technical Commission (IEC), the International Telecommunication Union (ITU), Underwriters Laboratories (UL) and the American National Standards Institute (ANSI). It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

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3.15 Port Damage. The input ports protection shall be achieved by either internal or external fusing. Capacitor and active devices shall be directly coupled to the terminals.

3.16 Connectors. The system shall have BNC & Banana type connectors available and comply with the total resistance and/or impedance stated in section 3.4 of this commercial item description.

3.17 Operating Temperature Range. The capacitance standard shall meet the specifications listed herein with full capabilities within the temperature range of 0°C to 50°C (32° to 122°F) after performing tests in MIL-PRF-28800 section 4.5.5.1.1 table 11 for class 3 equipment.

3.18 Storage Temperature Range. The capacitance standard shall meet and/or exceed manufacturer's specifications for stored equipment after tests of MIL-PRF-28800 section 4.5.5.1.1 table 11 for temperature range and relative humidity storage for class 3 equipment classification.

3.19 Altitude Requirements. The capacitance standard shall meet its operating performance and accuracy while operating at an altitude equal or greater than 2,440 meters (8,000 ft.). The equipment shall meet performance and accuracy requirements, for non-operating storage up to 3,050 meters (10,000 ft.).

3.20 Calibration & Maintenance Adjustments. No special tools or software shall be required for alignment or adjustment except those furnished with the capacitance standard test set.

3.21 Calibration Period. The design of the capacitance standard shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the instrument internal modules. The calibration interval shall be based on a period of one calendar year or an operating time of 2,080 hours or whichever occurs first.

3.22 Reliability. The design of the capacitance standard shall be such that under normal use and operation the mean time between failures (MTBF) exceeds the minimum of 2,048 hours of operation accordance with MIL-PRF-28800. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with MIL-PRF-28800 section. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding MIL-PRF-28800 sections and section.

3.23 Maintainability. The capacitance standard shall meet the maintainability criterion in accordance with MIL-STD-1472 paragraphs 5.9 through 5.9.18 and 5.9.21 and MIL-PRF-28800 sections 3.14.1 through 3.14.4 and sections 6.6.2.4 and 6.6.24.1.

3.24 Preventive Maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800 sections 3.14.2 & 3.14.3.

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3.25 Corrective Maintenance. The capacitance standard shall be designed to meet reparability criterion in accordance with section 3.14.3.

3.26 Electrical Power Sources & Connections. The capacitance standard is a passive device based instrument. The capacitance standard shall have a grounding strap and terminal for floating and grounded applications.

3.27 Dimensions. The capacitance standard shall comply with width and depth of EIA/ECA-310-E standard equipment rack configuration. The capacitance standard enclosure shall fit in a storage drawer of a 19" ANSI/EIA rack configuration. The enclosure shall be compatible with standard EIA storage of less than 5 U's. One unit or "U" is defined by EIA/ECA-310-E to be 1.75 inches or 44.45 mm.

3.28 Weight. The capacitance standard system shall comply with MIL-STD-1472G section 5.9.11.3.3 for weight & size restrictions for single person lift, carry or handling. The total weight of the capacitance standard, excluding accessories and manuals, shall not exceed a one-person lift of 4.5 kilograms, (10 pounds).

4. REGULATORY REQUIREMENTS.

4.1 Recycled Recovered Materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and the product conforms to the producer's engineering drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Source of Documents.

7.1.1 Military Specifications, Standards and Handbooks referenced herein may be obtained at https://assist.dla.,mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg. 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/.

7.1.3 ANSI/ASME documents may be obtained online at http://www.asme.org/ or from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

7.1.4 ISO documents can be obtained online at http://www.iso.org or from the International Organization for Standardization, ISO Central Secretariat, 1, ch de la Voie-Creuse, CP 56-CH-1211 Geneva 20, Switzerland.

7.1.5 EIA/ECA documents can be obtained online at http://www.eciaonline.org/default.aspx .

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7.2 Key Words.

Antenna
Cable
Frequency
Signal

Custodians:
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Preparing Activity:
Air Force – 84

Reviewers:
Air Force – 99

Agent:
Air Force – 99

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